In [2]:
```
import numpy as np
import pandas as pd

print(np.__version__)
print(pd.__version__)
```

```
2.5.4
3.1.0
```

In [2]:
```
import numpy as np
import pandas as pd

data = {
    "student_id": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "name": [
        "Arun", "Priya", "Rahul", "Sneha", "Kiran",
        "Divya", "Rohit", "Anjali", "Vijay", "Meena"
    ],
    "department": [
        "CSE", "ECE", "CSE", "IT", "ECE",
        "CSE", "IT", "ECE", "CSE", "IT"
    ],
    "subject": [
        "Python", "Java", "Python", "SQL", "Java",
        "SQL", "Python", "Java", "SQL", "Python"
    ],
    "marks": [85, 72, 91, 65, 78, 88, None, 95, 67, 82],
    "attendance": [92, 85, 98, 70, 88, 95, 60, 99, 75, 90],
    "study_hours": [4, 3, 5, 2, 3, 4, 1, 6, 2, 4],
    "gender": [
        "Male", "Female", "Male", "Female", "Male",
        "Female", "Male", "Female", "Male", "Female"
    ],
    "city": [
        "Chennai", "Hyderabad", "Bengaluru", "Chennai",
        "Hyderabad", "Bengaluru", "Chennai", None,
        "Bengaluru", "Chennai"
    ]
}

df = pd.DataFrame(data)

df
```

Out[2]:

	student_id	name	department	subject	marks	attendance	study_hours	gender	
0	101	Arun	CSE	Python	85.0	92	4	Male	C
1	102	Priya	ECE	Java	72.0	85	3	Female	Hyd
2	103	Rahul	CSE	Python	91.0	98	5	Male	Ben
3	104	Sneha	IT	SQL	65.0	70	2	Female	C
4	105	Kiran	ECE	Java	78.0	88	3	Male	Hyd
5	106	Divya	CSE	SQL	88.0	95	4	Female	Ben
6	107	Rohit	IT	Python	NaN	60	1	Male	C
7	108	Anjali	ECE	Java	95.0	99	6	Female	
8	109	Vijay	CSE	SQL	67.0	75	2	Male	Ben
9	110	Meena	IT	Python	82.0	90	4	Female	C

In [4]:
```
marks = np.array([10, 20, 30, 40, 50])
marks
```

Out[4]:
```
array([10, 20, 30, 40, 50])
```

In [5]:
```
type(marks)
```

Out[5]:
```
numpy.ndarray
```

In [6]:
```
print(marks.ndim)
print(marks.shape)
print(marks.size)
```

```
1
(5,)
5
```

In [5]:
```
print(marks.ndim)
print(marks.shape)
print(marks.size)
```

Out[5]:
```
numpy.ndarray
```

In [7]:
```
print(marks.min())
print(marks.max())
print(marks.sum())
print(marks.mean())
```

```
10
50
150
30.0
```

In [8]:
```
print(marks * 2)
print(marks + 5)
print(marks.reshape(5, 1))
```

```
[ 20  40  60  80 100]
[15 25 35 45 55]
[[10]
 [20]
 [30]
 [40]
 [50]]
```

In [9]:
```
print(marks[marks > 25])
print(marks[marks % 2 == 0])
```

```
[30 40 50]
[10 20 30 40 50]
```

In [ ]:
```

```

In [11]:
```
arr2 = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])

arr2
```

Out[11]:
```
array([[10, 20, 30],
       [40, 50, 60],
       [70, 80, 90]])
```

In [12]:
```
print(arr2.ndim)
print(arr2.shape)
print(arr2.size)
```

```
2
(3, 3)
9
```

In [13]:
```
print(arr2.sum(axis=1))
```

```
[ 60 150 240]
```

In [14]:
```
print(arr2.sum(axis=0))
```

```
[120 150 180]
```

In [15]:
```
print(arr2.sum())
print(arr2.mean())
```

```
450
50.0
```

In [16]:
```
print(arr2.min())
print(arr2.max())
```

```
10
90
```

In [ ]:

In [17]:
```
print(df.head())
```

```
   student_id   name department subject  marks  attendance  study_hours  \
0         101   Arun        CSE  Python   85.0          92            4
1         102  Priya        ECE    Java   72.0          85            3
2         103  Rahul        CSE  Python   91.0          98            5
3         104  Sneha         IT     SQL   65.0          70            2
4         105  Kiran        ECE    Java   78.0          88            3

   gender       city
0    Male    Chennai
1  Female  Hyderabad
2    Male  Bengaluru
3  Female    Chennai
4    Male  Hyderabad
```

In [18]:
```
print(df.head(3))
```

```
   student_id   name department subject  marks  attendance  study_hours  \
0         101   Arun        CSE  Python   85.0          92            4
1         102  Priya        ECE    Java   72.0          85            3
2         103  Rahul        CSE  Python   91.0          98            5

   gender       city
0    Male    Chennai
1  Female  Hyderabad
2    Male  Bengaluru
```

In [19]:
```
print(df.tail())
```

```
   student_id    name department subject  marks  attendance  study_hours  \
5         106   Divya        CSE     SQL   88.0          95            4
6         107   Rohit         IT  Python    NaN          60            1
7         108  Anjali        ECE    Java   95.0          99            6
8         109   Vijay        CSE     SQL   67.0          75            2
9         110   Meena         IT  Python   82.0          90            4

   gender       city
5  Female  Bengaluru
6    Male    Chennai
7  Female        NaN
8    Male  Bengaluru
9  Female    Chennai
```

In [20]:
```
df.tail(2)
```

Out[20]:

	student_id	name	department	subject	marks	attendance	study_hours	gender	
8	109	Vijay	CSE	SQL	67.0	75	2	Male	Beng
9	110	Meena	IT	Python	82.0	90	4	Female	Ch

In [21]:
```
df.shape
```

Out[21]:
```
(10, 9)
```

In [22]:
```
df.columns
```

Out[22]:
```
Index(['student_id', 'name', 'department', 'subject', 'marks', 'attendance',
       'study_hours', 'gender', 'city'],
      dtype='str')
```

In [23]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   student_id   10 non-null     int64
 1   name         10 non-null     str
 2   department   10 non-null     str
 3   subject      10 non-null     str
 4   marks        9 non-null      float64
 5   attendance   10 non-null     int64
 6   study_hours  10 non-null     int64
 7   gender       10 non-null     str
 8   city         9 non-null      str
dtypes: float64(1), int64(3), str(5)
memory usage: 852.0 bytes
```

In [24]:
```
df.describe()
```

Out[24]:

	student_id	marks	attendance	study_hours
count	10.00000	9.000000	10.000000	10.000000
mean	105.50000	80.333333	85.200000	3.400000
std	3.02765	10.606602	12.899612	1.505545
min	101.00000	65.000000	60.000000	1.000000
25%	103.25000	72.000000	77.500000	2.250000
50%	105.50000	82.000000	89.000000	3.500000
75%	107.75000	88.000000	94.250000	4.000000
max	110.00000	95.000000	99.000000	6.000000

In [25]:
```
df.describe(include="all")
```

Out[25]:

	student_id	name	department	subject	marks	attendance	study_hours	gend
count	10.00000	10	10	10	9.000000	10.000000	10.000000	
unique	NaN	10	3	3	NaN	NaN	NaN	
top	NaN	Arun	CSE	Python	NaN	NaN	NaN	Ma
freq	NaN	1	4	4	NaN	NaN	NaN	
mean	105.50000	NaN	NaN	NaN	80.333333	85.200000	3.400000	Na
std	3.02765	NaN	NaN	NaN	10.606602	12.899612	1.505545	Na
min	101.00000	NaN	NaN	NaN	65.000000	60.000000	1.000000	Na
25%	103.25000	NaN	NaN	NaN	72.000000	77.500000	2.250000	Na
50%	105.50000	NaN	NaN	NaN	82.000000	89.000000	3.500000	Na
75%	107.75000	NaN	NaN	NaN	88.000000	94.250000	4.000000	Na
max	110.00000	NaN	NaN	NaN	95.000000	99.000000	6.000000	Na

In [26]:
```
df.dtypes
```

Out[26]:
```
student_id       int64
name               str
department         str
subject            str
marks          float64
attendance       int64
study_hours      int64
gender             str
city               str
dtype: object
```

In [27]:
```
df.isnull()
```

Out[27]:

	student_id	name	department	subject	marks	attendance	study_hours	gender	city
0	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False	False	False
6	False	False	False	False	True	False	False	False	False
7	False	False	False	False	False	False	False	False	True
8	False	False	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False	False	False

In [28]:
```
df.isnull().sum()
```

Out[28]:
```
student_id     0
name           0
department     0
subject        0
marks          1
attendance     0
study_hours    0
gender         0
city           1
dtype: int64
```

In [29]:
```
df[df.isnull().any(axis=1)]
```

Out[29]:

	student_id	name	department	subject	marks	attendance	study_hours	gender	c
6	107	Rohit	IT	Python	NaN	60	1	Male	Chen
7	108	Anjali	ECE	Java	95.0	99	6	Female	N

In [32]:
```
df["marks"].isnull().sum()
```

Out[32]:
```
np.int64(1)
```

In [33]:
```
df.duplicated().sum()
```

Out[33]:
```
np.int64(0)
```

In [34]:
```
df[df.duplicated()]
```

Out[34]:

student_id	name	department	subject	marks	attendance	study_hours	gender	city

In [35]:
```
df = df.drop_duplicates()
```

In [36]:
```
df["study_score"] = df["marks"] * df["study_hours"]

print(df[["name", "marks", "study_hours", "study_score"]])
```

```
     name  marks  study_hours  study_score
0    Arun   85.0            4        340.0
1   Priya   72.0            3        216.0
2   Rahul   91.0            5        455.0
3   Sneha   65.0            2        130.0
4   Kiran   78.0            3        234.0
5   Divya   88.0            4        352.0
6   Rohit    NaN            1          NaN
7  Anjali   95.0            6        570.0
8   Vijay   67.0            2        134.0
9   Meena   82.0            4        328.0
```

In [37]:
```
df
```

Out[37]:

	student_id	name	department	subject	marks	attendance	study_hours	gender	
0	101	Arun	CSE	Python	85.0	92	4	Male	C
1	102	Priya	ECE	Java	72.0	85	3	Female	Hyd
2	103	Rahul	CSE	Python	91.0	98	5	Male	Ben
3	104	Sneha	IT	SQL	65.0	70	2	Female	C
4	105	Kiran	ECE	Java	78.0	88	3	Male	Hyd
5	106	Divya	CSE	SQL	88.0	95	4	Female	Ben
6	107	Rohit	IT	Python	NaN	60	1	Male	C
7	108	Anjali	ECE	Java	95.0	99	6	Female	
8	109	Vijay	CSE	SQL	67.0	75	2	Male	Ben
9	110	Meena	IT	Python	82.0	90	4	Female	C

In [38]:
```
df["marks_bonus"] = df["marks"] + 5

print(df[["name", "marks", "marks_bonus"]])
```

```
     name  marks  marks_bonus
0    Arun   85.0         90.0
1   Priya   72.0         77.0
2   Rahul   91.0         96.0
3   Sneha   65.0         70.0
4   Kiran   78.0         83.0
5   Divya   88.0         93.0
6   Rohit    NaN          NaN
7  Anjali   95.0        100.0
8   Vijay   67.0         72.0
9   Meena   82.0         87.0
```

In [39]:
```
df["marks"] = df["marks"].fillna(df["marks"].mean())
df["city"] = df["city"].fillna("Unknown")
print(df.isnull().sum())
```

```
student_id     0
name           0
department     0
subject        0
marks          0
attendance     0
study_hours    0
gender         0
city           0
study_score    1
marks_bonus    1
dtype: int64
```

In [41]:
```
df["study_score"] = df["marks"] * df["study_hours"]
df["marks_bonus"] = df["marks"] + 5
```

In [42]:
```
print("Total marks:", df["marks"].sum())
print("Average marks:", df["marks"].mean())
print("Minimum marks:", df["marks"].min())
print("Maximum marks:", df["marks"].max())
print("Number of students:", df["marks"].count())

print("Average attendance:", df["attendance"].mean())
print("Total study hours:", df["study_hours"].sum())
```

```
Total marks: 803.3333333333333
Average marks: 80.33333333333333
Minimum marks: 65.0
Maximum marks: 95.0
Number of students: 10
Average attendance: 85.2
Total study hours: 34
```

In [43]:
```
print(df.groupby("department")["marks"].mean())
print(df.groupby("department")["marks"].sum())
print(
    df.groupby("department")["marks"]
      .agg(["sum", "mean", "min", "max", "count"])
)
```

```
department
CSE    82.750000
ECE    81.666667
IT     75.777778
Name: marks, dtype: float64
department
CSE    331.000000
ECE    245.000000
IT     227.333333
Name: marks, dtype: float64
                  sum       mean   min   max  count
department
CSE         331.000000  82.750000  67.0  91.0      4
ECE         245.000000  81.666667  72.0  95.0      3
IT          227.333333  75.777778  65.0  82.0      3
```

In [44]:
```
print(df.groupby("subject")["marks"].mean())
print(df.groupby("subject")["marks"].sum())
```

```
subject
Java      81.666667
Python    84.583333
SQL       73.333333
Name: marks, dtype: float64
subject
Java      245.000000
Python    338.333333
SQL       220.000000
Name: marks, dtype: float64
```

In [45]:
```
print(df[df["marks"] > 80])
print(df[df["marks"] < 70])
print(df[df["marks"] == 95])
print(df[df["attendance"] >= 90])
print(df[
    (df["marks"] > 80) &
    (df["attendance"] > 90)
])
print(df[
    (df["marks"] > 90) |
    (df["attendance"] < 70)
])
print(df[~(df["department"] == "CSE")])
```

```
   student_id    name department subject      marks  attendance  study_hours  \
0         101    Arun        CSE  Python  85.000000          92            4
2         103   Rahul        CSE  Python  91.000000          98            5
5         106   Divya        CSE     SQL  88.000000          95            4
6         107   Rohit         IT  Python  80.333333          60            1
7         108  Anjali        ECE    Java  95.000000          99            6
9         110   Meena         IT  Python  82.000000          90            4

   gender       city  study_score  marks_bonus
0    Male    Chennai   340.000000    90.000000
2    Male  Bengaluru   455.000000    96.000000
5  Female  Bengaluru   352.000000    93.000000
6    Male    Chennai    80.333333    85.333333
7  Female    Unknown   570.000000   100.000000
9  Female    Chennai   328.000000    87.000000
   student_id   name department subject  marks  attendance  study_hours  \
3         104  Sneha         IT     SQL   65.0          70            2
8         109  Vijay        CSE     SQL   67.0          75            2

   gender       city  study_score  marks_bonus
3  Female    Chennai        130.0         70.0
8    Male  Bengaluru        134.0         72.0
   student_id    name department subject  marks  attendance  study_hours  \
7         108  Anjali        ECE    Java   95.0          99            6

   gender     city  study_score  marks_bonus
7  Female  Unknown        570.0        100.0
   student_id    name department subject  marks  attendance  study_hours  \
0         101    Arun        CSE  Python   85.0          92            4
2         103   Rahul        CSE  Python   91.0          98            5
5         106   Divya        CSE     SQL   88.0          95            4
7         108  Anjali        ECE    Java   95.0          99            6
9         110   Meena         IT  Python   82.0          90            4

   gender       city  study_score  marks_bonus
0    Male    Chennai        340.0         90.0
2    Male  Bengaluru        455.0         96.0
5  Female  Bengaluru        352.0         93.0
7  Female    Unknown        570.0        100.0
9  Female    Chennai        328.0         87.0
   student_id    name department subject  marks  attendance  study_hours  \
0         101    Arun        CSE  Python   85.0          92            4
2         103   Rahul        CSE  Python   91.0          98            5
5         106   Divya        CSE     SQL   88.0          95            4
7         108  Anjali        ECE    Java   95.0          99            6

   gender       city  study_score  marks_bonus
0    Male    Chennai        340.0         90.0
2    Male  Bengaluru        455.0         96.0
5  Female  Bengaluru        352.0         93.0
7  Female    Unknown        570.0        100.0
   student_id    name department subject      marks  attendance  study_hours  \
2         103   Rahul        CSE  Python  91.000000          98            5
6         107   Rohit         IT  Python  80.333333          60            1
7         108  Anjali        ECE    Java  95.000000          99            6
```

```
   gender       city  study_score  marks_bonus
2    Male  Bengaluru   455.000000    96.000000
6    Male    Chennai    80.333333    85.333333
7  Female    Unknown   570.000000   100.000000
   student_id    name department subject      marks  attendance  study_hours  \
1         102   Priya        ECE    Java  72.000000          85            3
3         104   Sneha         IT     SQL  65.000000          70            2
4         105   Kiran        ECE    Java  78.000000          88            3
6         107   Rohit         IT  Python  80.333333          60            1
7         108  Anjali        ECE    Java  95.000000          99            6
9         110   Meena         IT  Python  82.000000          90            4

   gender       city  study_score  marks_bonus
1  Female  Hyderabad   216.000000    77.000000
3  Female    Chennai   130.000000    70.000000
4    Male  Hyderabad   234.000000    83.000000
6    Male    Chennai    80.333333    85.333333
7  Female    Unknown   570.000000   100.000000
9  Female    Chennai   328.000000    87.000000
```

In [46]:
```
df["result"] = np.where(
    df["marks"] >= 40,
    "Pass",
    "Fail"
)

print(df[["name", "marks", "result"]])
```

```
     name      marks result
0    Arun  85.000000   Pass
1   Priya  72.000000   Pass
2   Rahul  91.000000   Pass
3   Sneha  65.000000   Pass
4   Kiran  78.000000   Pass
5   Divya  88.000000   Pass
6   Rohit  80.333333   Pass
7  Anjali  95.000000   Pass
8   Vijay  67.000000   Pass
9   Meena  82.000000   Pass
```

In [ ]: